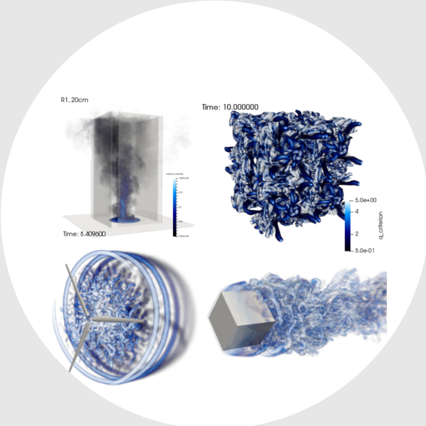



# Stefan P. Domino, Ph.D.

Computational Scientist

COMERI
PO Box 1259
Sandia Park, NM, 87047-1259

spdomino@comeri.org
spdomin@stanford.edu

SPD's Stanford Profile Page


## What I Do

I transform people's understanding of the world by deploying high-performing computational fluid dynamics tools. While some draw analogy to such computational tools as the rasp in Michelangelo's hand, I choose to view the partnership analogous to that of the luthier and the violinist in that both are required to make new that which was formerly unheard.

## Skills/Attributes

low-Mach Fluids

Turbulence

Next-generation-platforms

Software Development

Extrovert

Driven

Passionate

[Scale ranges from 0 (Fundamental Awareness) to 6 (Expert).]

## Interests

My professional interest resides in the development and deployment of computational fluid dynamics (CFD) tools to facilitate a transformative understanding of otherwise intractable physical phenomena. Exercising these tools in partnership with theory and experiments provides illumination of coupled physics processes. Modeling low-Mach multi-physics applications that include turbulence (generally large-eddy simulation and direct numerical simulation-based), variable-density buoyancy effects, multiphase, and chemical reactions can reveal extraordinarily complex fluids, thermal, and species structures thereby allowing that which is generally unseen to be fully appreciated. Fostering partnerships within a high-performing team to solve grand-challenge problems, in addition to novel usages of SciML/AI is desired.

## Current Position(s) Held

COMERI — CEO/President/Esteemed Technical Staff, Computational Marine Ethology Research Institute.

Stanford — Adjunct Professor; School of Engineering, Institute for Computational and Mathematical Engineering (ICME).

## Education

**2000** — Doctor of Philosophy — Chemical Engineering, University of Utah
Researched and deployed advanced modeling and simulation techniques to more accurately predict the oxides of nitrogen ($NO_x$) in multiphase combustion applications. Advisor: Professor Philip Smith

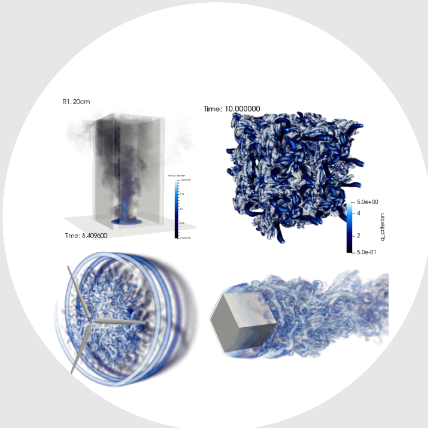
**1994** — Bachelor of Science — Chemical Engineering, University of Utah
Researched the use of per-fluorocarbons for advanced mammalian bioreactor design. Advisor: Professor Edward Trujillo.

## Recent Experience

**2018-now** — CEO/President/Esteemed Technical Staff — COMERI
Lead management, research, and funding objectives for the Computational Marine Ethology Research Institute (COMERI) - a 501(c)(3) nonprofit research Institute that drives foundational understanding of marine ethology using first-principles physics.

**2021-now** — Adjunct Professor — Stanford/ICME
Teaching responsibilities for Stanford's ME469 Mechanical Engineering graduate *Computational Methods for Fluid Dynamics* course where Mare-Nalu is used as pedagogical tool to bridge foundational numerical methods development and practical production CFD; support mentoring of graduate students and post-doctoral appointees.

**2000-2025** — Technical Staff — Sandia National Laboratories
Extreme scale, low-Mach turbulent fluid mechanics methods development for complex systems that drive the coupling of mass, momentum, species and energy transport. As PI, my research resides within the intersection of physics model development, numerical methods research, V&V techniques exploration, SciML/AI, and high-performance computing and coding methods for low-Mach flow; originator of the open-source Nalu code base. I am proud to have served the Lab's response to National crises such as Deep Water Horizon and the COVID-19 pandemic. Served as PI and lead developer for the generally unstructured, massively parallel Sierra/Fuego code base and was a team contributor to the NNSA Defense Programs Awards of Excellence for significant contributions Stockpile Stewardship Program. Promotion to Senior Member of the Technical Staff from post-doc (2001), Principal Member of the Technical Staff (2005), and Distinguished Member of the Technical Staff (2022).



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## Goals

My primary career goal centers on extending state-of-the art in CFD methods to facilitate the advanced deployment of credible tools that support a wide range of atypical, e.g., computational marine ethology, fire, wind/wave-energy – with an emphasis on multi-physics applications. More recently, I have been involved in SciML/Ai-based method for the advancement of scientific understandings and have completely transitioned to COMERI-based research themes. Mentoring, teaching, and motivating the next generation of computational scientists captures my core passion.

## Investment Areas

SciML-AI

Next-Generation Platforms

Wave Energy

VVUQ

low-Mach

Turbulence

Marine Ethology

[Scale ranges from 0 (low-interest) to 6 (high-interest).]

## Recent Peer-reviewed Publications

2025 Domino, S. P., Scott, S., Hubbard, J., *Structural uncertainty assessment for fire-engulfed objects in crosswind: Establishing credibility for a multiphysics wall-modeled large-eddy simulation paradigm*, Phys. Rev. Fluids, https://doi.org/10.1103/PhysRevFluids.10.054604.

2024 Domino, S. P. *On the subject of large-scale pool fires and turbulent boundary layer interactions*, Phys. Fluids, https://doi.org/10.1063/5.0196265.

2023 Domino, S. P., Wenzel, E., *A direct numerical simulation study for confined non-isothermal jet impingement at moderate nozzle-to-plate distances: capturing jet-to-ambient density effects*, Int. J. Heat Mass Trans, https://doi.org/10.1016/j.ijheatmasstransfer.2023.124168.

2022 Scott, S. N., Domino, S. P., *A computational examination of large-scale pool fires: variations in crosswind velocity and pool shape*, Flow, https://doi.org/10.1017/flo.2022.26.

2022 Domino, S. P., Horne, W., *Development and deployment of a credible unstructured, six-DOF, implicit low-Mach overset simulation tool for wave energy applications*, Renewable Energy, https://doi.org/10.1016/j.renene.2022.09.005.

2021 Domino, S. P., Hewson, J., Knaus, R., Hansen, M., *Predicting large-scale pool fire dynamics using an unsteady flamelet- and large-eddy simulation-based model suite*, Phys. Fluids, https://doi.org/10.1063/5.0060267 (Editor's pick).

2021 Domino, S. P., *A case study on pathogen transport, deposition, evaporation and transmission: linking high-fidelity computational fluid dynamics simulations to probability of infection*, Int. J. CFD, https://doi.org/10.1080/10618562.2021.1905801.

2021 Domino, S. P., Pierce, F., Hubbard, J., *A multi-physics computational investigation of droplet pathogen transport emanating from synthetic coughs and breathing*, Atom. Sprays, https://doi.org/10.1615/AtomizSpr.2021036313.

2019 Jofre, L., Domino, S. P., Iaacarino, G., *Eigensensitivity analysis of subgrid-scale stresses in large-eddy simulation of a turbulent axisymmetric jet*, Int. J. Heat Fluid Flow, https://doi.org/DOI:10.1016/J.IJHEATFLUIDFLOW.2019.04.014.

2019 Domino, S. P., Sakievich, P., Barone, M., *An assessment of atypical mesh topologies for low-Mach large-eddy simulation*, Comp. Fluids, https://doi.org/10.1016/j.compfluid.2018.12.002.

2018 Domino, S. P., *Design-order, non-conformal low-Mach fluid algorithms using a hybrid CVFEM/DG approach*, J. Comput. Phys., https://doi.org/10.1016/j.jcp.2018.01.007.

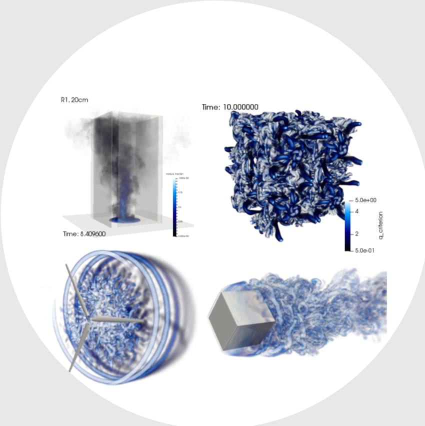
2018 Jofre, L., Domino, S. P., Iaacarino, G., *A framework for characterizing structural uncertainty in large-eddy simulation closures*, Flow Turb. Combust., https://doi.org/10.1007/s10494-017-9844-8.

## Noteworthy Publications

2023 Benjamin, M., Domino, S. P., Iaccarino, G., *Neural networks for large eddy simulations of wall-bounded turbulence: numerical experiments and challenges*, Eur. Phys. J. E, https://doi.org/10.1140/epje/s10189-023-00314-6.

2022 Barone, M., Ray, J., Domino, S. P., *Feature selection, clustering, and prototype placement for turbulence datasets"*, AIAA J., https://doi.org/10.2514/1.J060919.

2014 Lin, P., Bettencourt, M., Domino, S. P., et al., *Towards extreme-scale simulations for low-Mach fluids with second-generation Trilinos*, Parallel Processing Letters, https://doi.org/10.1142/S0129626414420055.



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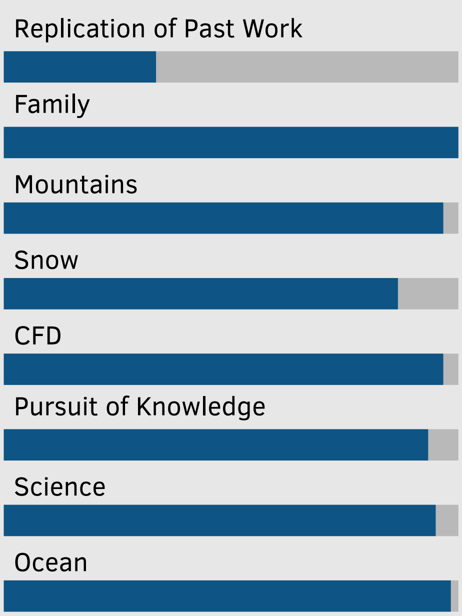
## Why I do it

The ability to explore multiphysics applications from a foundational modeling and simulation perspective is critical to future scientific advances. This high-level motivation has driven my desire to work within the intersection of physics elucidation, numerical methods development, and code development. More recently, the ability to deploy advanced uncertainty quantification (UQ) techniques to drive physics understanding, which may include structural uncertainty methods, machine learning approaches, etc., has transformed the former research paradigm.

## Favorite Things




[Scale ranges from 0 (unfavorable) to 6 (favorable).]

## Notable Projects as PI

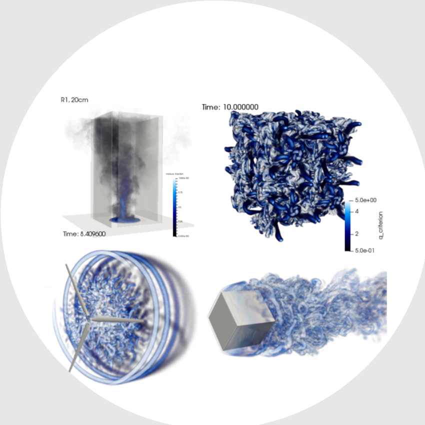
| | |
|---|---|
| 2020-2025 | *Agile Physics and Engineering Models*. Advanced thermal/thermal radiation coupling techniques; elucidation of non-isothermal jet impingement physics; WM-RLES, & data science turbulence modeling. |
| 2019-2025 | *VVUQ Methods for Turbulent Flow*. Exploring fire-engulfed objects subjected to croswind dynamics in accident scenarios; developing structural uncertainties for LES through eigenvalue decomposition and perturbation of stresses towards limiting turbulence states. |
| 2021-2022 | *Developing credible high-fidelity mod/sim tools for wave energy converter design*. Development of implicit overset methods coupled to six-DOF and volume of fluid transport. |
| 2020-2021 | *COVID-19 Transportation and Transmission*. High-fidelity pathogen modeling approaches for breathing and coughing events that allows for the ability to distinguish between droplets that deposit and those that form persisting aerosol plumes. |
| 2012-2015 | *Computer Science Advanced Research: Core Computational Methodologies*. Portfolio manager for a multi-discipline Advanced Simulation and Computing (Research Foundations) included funding decisions ($1.25 million per year) and technical oversight of advanced methods of algebraic multigrid, the development of Helmholtz solvers, etc. |
| 2003 - 2006 | *Sierra/Fuego Integrated Codes Project*. Fuego is the Sandia National Laboratories turbulent reacting flagship fire physics simulation tool that supports Science-based Stockpile Stewardship. |

## Distinguished Awards

| | |
|---|---|
| 2017 | Sheldon R. Tieszen Sandia National Labs Engineering Sciences Award for a distinguished career in pursuit of technical excellence. |

## Noteworthy Presentations

| | |
|---|---|
| 2025 | Domino, S. P., *Structural uncertainty assessment for fire-engulfed objects in crosswind: Establishing credibility for a multiphysics wall-modeled large-eddy simulation paradigm*, Physical Review Journal Club. |
| 2023 | Domino, S. P., *Building a Credible, Open-Source High-Fidelity Computational Fluid Dynamics Tool Suitable for Renewable Wave Energy Applications*, Invited Stanford CTR Tea Seminar Series. |
| 2022 | Domino, S. P., *Exploring high-fidelity computational fluid dynamics approaches for airborne pandemic risk mitigation*, Invited Stanford CTR Tea Seminar Series. |
| 2021 | Domino, S. P., *A historical perspective on Sandia National Laboratories fire science mod/sim philosophy: The role of high performance computing and unstructured numerical methods advances*, Invited University of Utah Graduate Seminar Series. |
| 2020 | Domino, S. P., *An evolution of a mindset: A historical perspective on Sandia National Laboratories Fire Science philosophy*, Invited Stanford CTR Tea Seminar Series. |
| 2018 | Domino, S. P., *Multi-phase use cases within the abnormal thermal environment*, Invited PSAAP-2 Multi-phase Workshop. |
| 2018 | Domino, S. P., *The suitability of hybrid meshes for low-Mach large-eddy simulation*, Invited LLNL/CASC Seminar Series. |
| 2017 | Domino, S. P., *ECP ExaWind experience in transitioning Nalu to MPI+x*, Invited PSAAP-2 Review. |
| 2009 | Domino, S. P., *Computational approaches to multiphysics applications: Predicting an object's thermal response within a turbulent reacting, participating media radiation environment*, Plenary Invitation, SIAM Conference on Computational Science and Engineering. |



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## Book Chapters

2022 Domino, S. P., *Unstructured finite volume approaches for turbulence.*, In: Moser, R. (ed), in Numerical Methods in Turbulence Simulation, Academic Press.

2017 Eldred, M., Ng, A., Barone, M., Domino, S. P., *Multifidelity uncertainty quantification using spectral stochastic discrepancy models*, In: Ghanem R., Higdon D., Owhadi H. (eds) in Handbook of Uncertainty Quantification., Springer.

## Select Sandia National Laboratories Reports

2025 Domino, S. P., *Structural uncertainty assessment for low-Mach wall-resolved large-eddy simulations: Plane Channel and Periodic Hill Use Cases*, Sandia National Laboratories, Sandia Report SAND2025-11249.

2019 Domino, S. P., Ananthan, S., Knaus, R., Williams, A., *Deploying Nalu/Kokos algorithmic infrastructure with performance benchmarking*, Sandia National Laboratories, Sandia Report SAND2017-10549R.

2015 Domino, S. P., *Sierra Low Mach Module: Nalu Theory Manual 1.0*, Sandia National Laboratories, SAND2015-3107W.

2005 Nicolette, V., Tieszen, S., Black, A., Domino, S. P., O'Hern, T., *A turbulence model for buoyant flows based on vorticity generation*, Sandia National Laboratories, Sandia Report SAND2005-6273.

## Past Preparation

2000-2001 Postdoctoral appointee — Sandia National Laboratories
Development of a smoke transport simulation tool for cargo bay fires in support of the FAA's response to ValueJet Flight 592. This work was recognized as part of the NASA Associate Administrator's Choice Award for Outstanding Accomplishment (Glenn Research Center) and a R&D 100 Award for the development of a multi-parameter, micro-sensor-based low false alarm fire detection system.

## Noteworthy Experiences

2020 - now Teaching Me469, Stanford Mechanical Engineering Department Graduate course (2023-2025); co-teaching 2020 & 2021: *Computational Methods for Fluid Dynamics*.

2006-2024 Multiple visiting scholar at Stanford's CTR Summer Program.

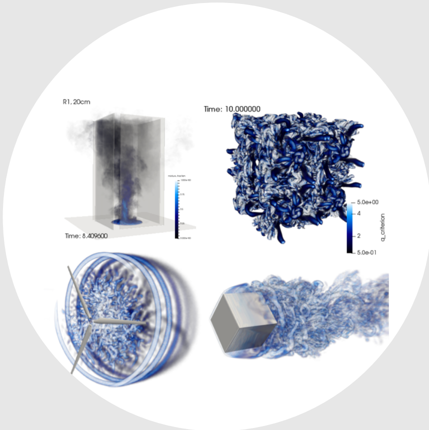
2001-now Mentoring of multiple post-doctoral researchers and graduate students; numerous internal and external peer-reviews supported including journals, DOE panels, NSF, Swiss NSF, and others.

## Miscellaneous Papers

2022 Hubbard, J., Hansen, M., Kirsch, J., Hewson, J., Domino, S. P., *Medium scale methanol pool fire model validation*, J. Heat Transfer, https://doi.org/10.1115/1.4054204.

2022 Hubbard, J., Cheng, M.D., Domino, S. P., *Mixing in low Reynolds number reacting impinging jets in crossflow*, J. Fluids. Engr., https://doi.org/10.1115/1.4056894

2000 Domino, S. P., Smith, P. J., *State space sensitivity to a prescribed probability density function shape in coal combustion systems: Joint $\beta$-PDF versus clipped Gaussian PDF*, Proc. Combust. Inst., https://doi.org/10.1016/S0082-0784(00)80644-X.



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## Stanford Center for Turbulence (CTR) Briefs

2024 Domino, S. P., *Credibility assessment for WMLES that includes fire-engulfed objects in crosswind*, Proceedings of the CTRSP (Cover image).

2022 Benjamin, M., Domino, S. P., Iaccarino, G., *Challenges in the use of neural networks for large–eddy simulations*, Annual CTR Research Briefs.

2018 Domino, S. P., Jofre, L., Iaccarino, G., *The suitability of hybrid meshes for low-Mach large-eddy simulation*, Proceedings of the CTR Summer Porogram (CTPSP). Jofre, L., Domino, S. P., Iaccarino, G., *Characterization of structural uncertainty in LES of a round jet*, Annual CTR Research Briefs.

2016 Domino, S. P., Jofre, L., Iaccarino, G., *Exploring model-form uncertainties in large-eddy simulation*, Proceedings of the CTRSP. Jofre, L., Domino, S. P., Iaccarino, G., *A framework for estimating uncertainty in LES closures*, Annual CTR Research Briefs

2014 Domino, S. P., *A comparison between low-order and higher-order low-Mach discretization approaches*, Proceedings of the CTRSP.

2010 Domino, S. P., *Towards verification of sliding mesh algorithms for complex applications using MMS*, Proceedings of the CTRSP.

2008 Domino, S. P., *A comparison of various equal-order interpolation methodologies using the method of manufactured solutions*, Proceedings of the CTRSP.

2006 Domino, S. P., *Toward verification of formal time accuracy for a family of approximate projection methods using the method of manufactured solutions*, Proceedings of the CTRSP.

## Select Conference Papers

2007 Domino, S. P., Wagner, G., Luketa-Hanlin, A., Black, A., Sutherland, J., *Verification for multi-mechanics applications*, $48^{th}$ AAIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference.

2002 Domino, S. P., Moen, C., Burns, S., Evans, G., *SIERRA/Fuego: A multi-mechanics fire environment simulation tool*, $41^{st}$ Aerospace Sciences Meeting and Exhibit.

2002 Domino, S. P., DesJardin, P., Suo-Antilla, J., *Development of a smoke transport model to enhance the certification process for cargo bay smoke detection systems*, Fire Safety Science Proceedings of the Seventh International Symposium.

## References

Please contact me for a comprehensive list of references.

## Review

Dr. Stefan Domino is a computational domain specialist researcher who develops tools and techniques to support advancement of multiphysics understanding of complex phenomena including turbulent fluid mechanics, heat transfer, and chemical reactions. Specific research thrusts center on high-fidelity computational modeling and simulation approaches for computational marine ethology, fire, wave- & wind-energy with a developing research thrust on generative AI, sustainability, and how computational science can support advanced approaches.